```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

df = pd.read_csv("/content/Mall_Customers")


print(df.head())
print(df.tail())
print(df.shape)
print(df.dtypes)
print(df.isnull().sum())
print("Duplicate records:", df.duplicated().sum())
print(df.describe())

X = df[["Annual Income (k$)", "Spending Score (1-100)"]]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

plt.figure(figsize=(8, 5))
plt.scatter(X["Annual Income (k$)"], X["Spending Score (1-100)"])
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.title("Annual Income vs Spending Score")
plt.show()

sns.pairplot(X)
plt.show()

inertia = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)

plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), inertia, marker="o")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.show()

k = 5

kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
df["Cluster"] = kmeans.fit_predict(X_scaled)

df.to_csv("Mall_Customers_Clustered.csv", index=False)

centroids_scaled = kmeans.cluster_centers_
centroids = scaler.inverse_transform(centroids_scaled)
```

```
plt.figure(figsize=(10, 6))
plt.scatter(
    df["Annual Income (k$)"],
    df["Spending Score (1-100)"],
    c=df["Cluster"],
    cmap="viridis",
    s=60
)

plt.scatter(
    centroids[:, 0],
    centroids[:, 1],
    c="red",
    marker="X",
    s=250,
    label="Centroids"
)

for i, (x, y) in enumerate(centroids):
    plt.text(x, y, f"C{i}: ({x:.2f}, {y:.2f})", fontsize=10)

plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.title("Customer Segmentation using K-Means")
plt.legend()
plt.show()

cluster_counts = df["Cluster"].value_counts().sort_index()
print("Number of customers in each cluster:")
print(cluster_counts)

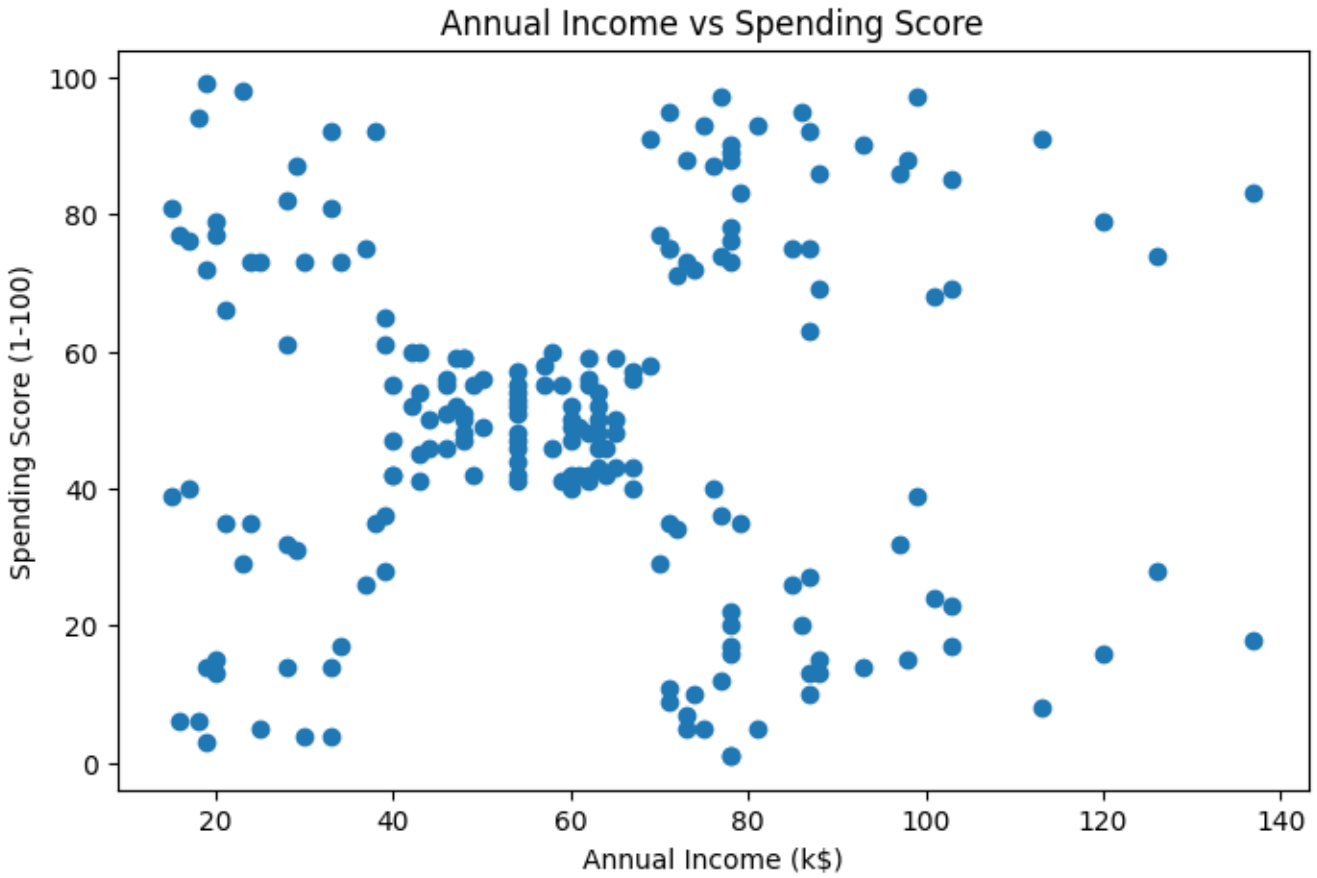
cluster_summary = df.groupby("Cluster")[["Annual Income (k$)", "Spending Score (1-100)"]].mean()

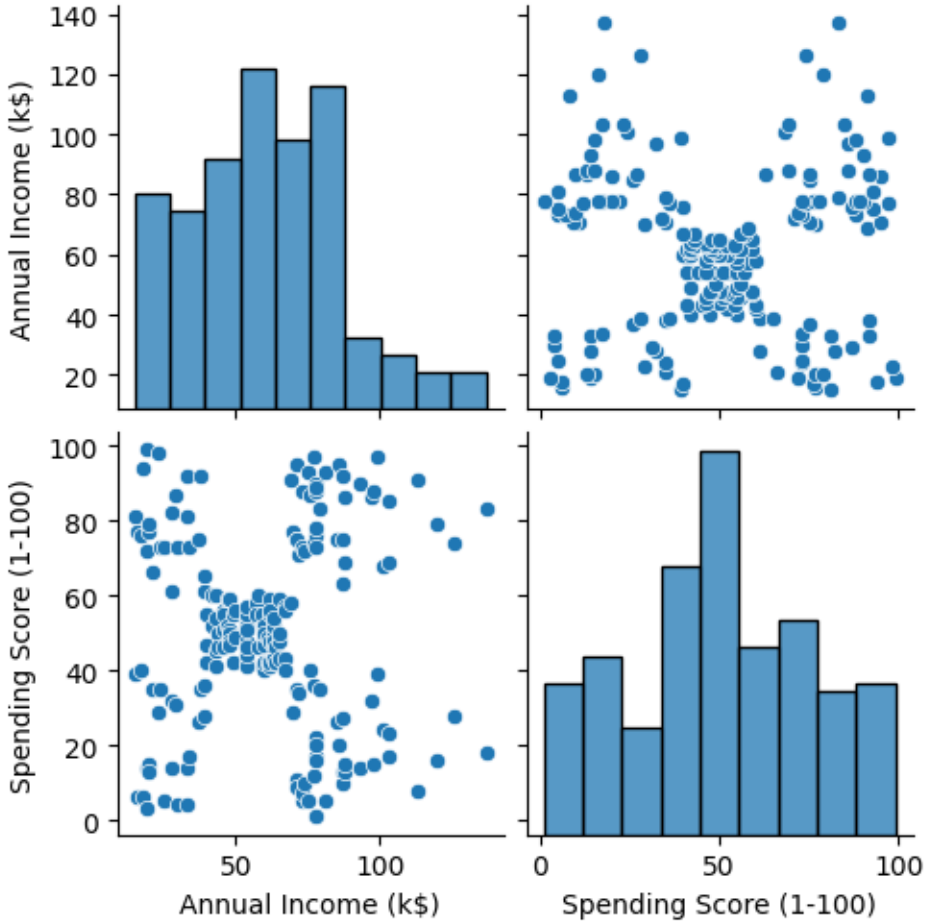
print("Average values for each cluster:")
print(cluster_summary)

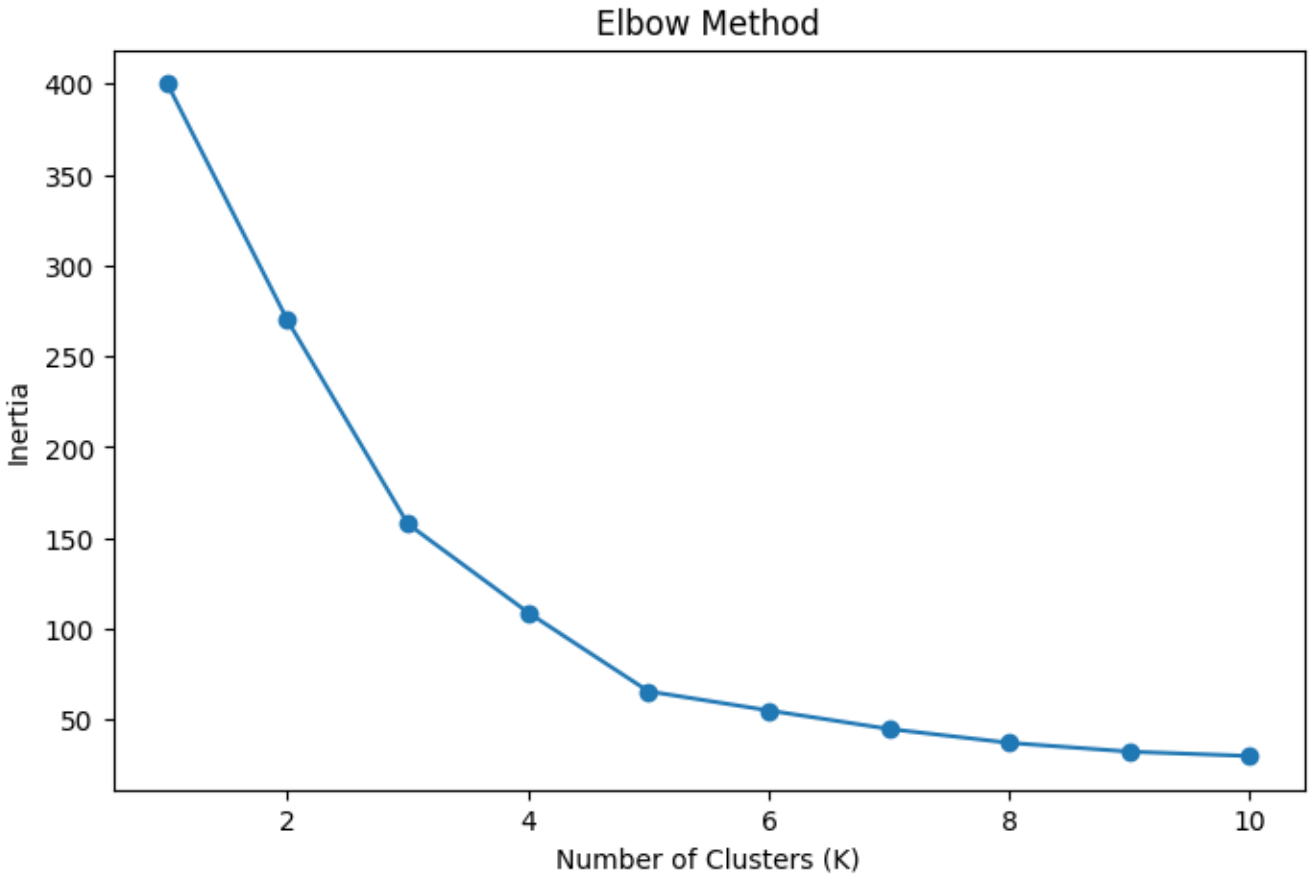
print("Cluster Centroids:")
for i, (income, score) in enumerate(centroids):
    print(f"Cluster {i}: Annual Income = {income:.2f}, Spending Score = {score:.2f}")

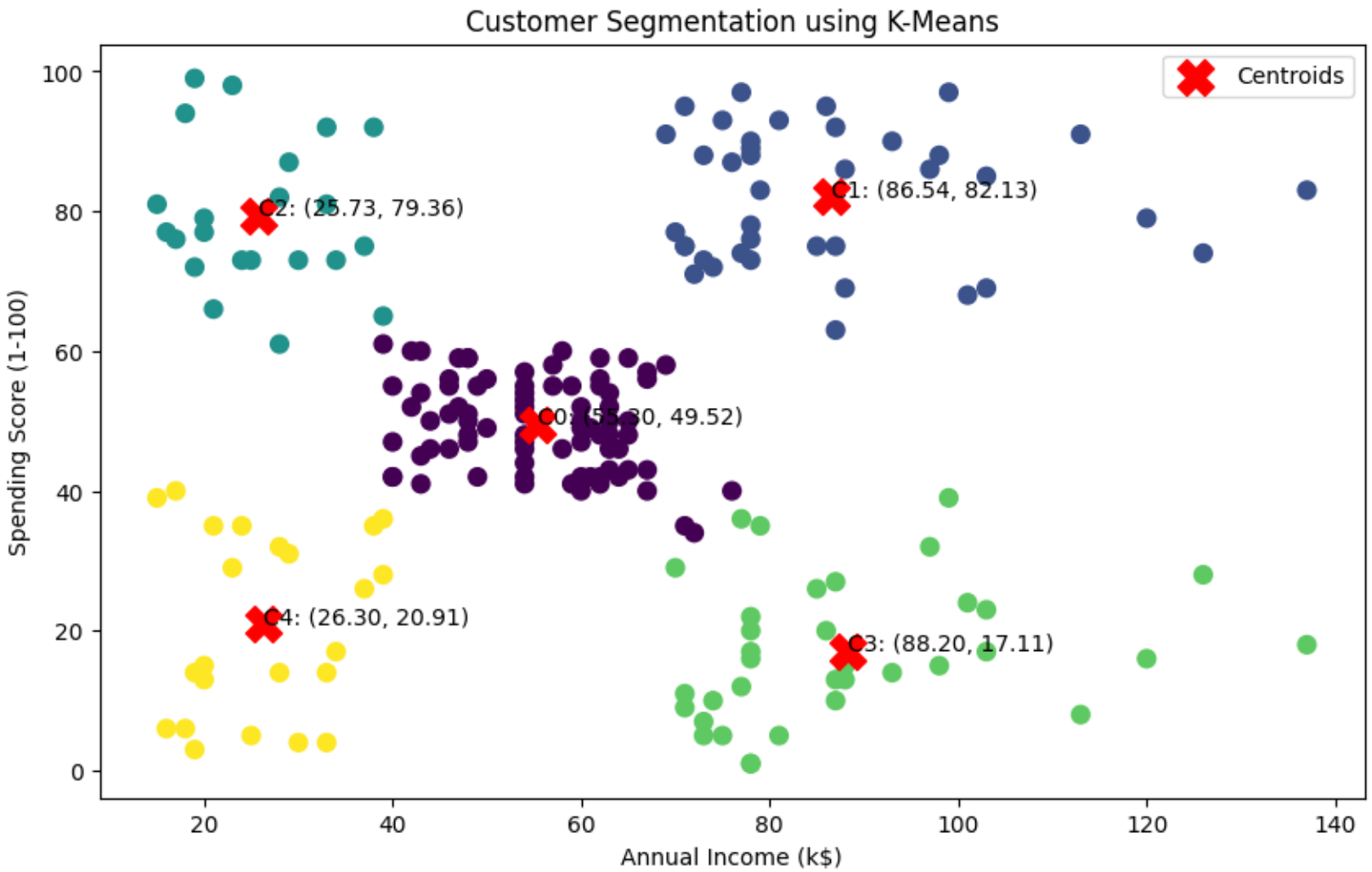
print("Clustered Dataset:")
print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
     CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
195         196  Female   35                 120                      79
196         197  Female   45                 126                      28
197         198    Male   32                 126                      74
198         199    Male   32                 137                      18
199         200    Male   30                 137                      83
(200, 5)
CustomerID                int64
Gender                   object
Age                       int64
Annual Income (k$)        int64
Spending Score (1-100)    int64
dtype: object
CustomerID              0
Gender                  0
Age                     0
Annual Income (k$)      0
Spending Score (1-100)  0
dtype: int64
Duplicate records: 0
       CustomerID         Age  Annual Income (k$)  Spending Score (1-100)
count  200.000000  200.000000          200.000000              200.000000
mean   100.500000   38.850000           60.560000               50.200000
std     57.879185   13.969007           26.264721               25.823522
min      1.000000   18.000000           15.000000                1.000000
25%     50.750000   28.750000           41.500000               34.750000
50%    100.500000   36.000000           61.500000               50.000000
75%    150.250000   49.000000           78.000000               73.000000
max    200.000000   70.000000          137.000000               99.000000
```


Annual Income vs Spending Score
Spending Score (1-100)
Annual Income (k$)
0
20
40
60
80
100
20
40
60
80
100
120
140


Annual Income (k$)
Spending Score (1-100)
Annual Income (k$)
Spending Score (1-100)
140
120
100
80
60
40
20
0
50
100


Elbow Method
Inertia
400
350
300
250
200
150
100
50
2
4
6
8
10
Number of Clusters (K)




```
Number of customers in each cluster:
Cluster
0    81
1    39
2    22
3    35
4    23
Name: count, dtype: int64
Average values for each cluster:
         Annual Income (k$)  Spending Score (1-100)
Cluster
```

```
0                55.296296              49.518519
1                86.538462              82.128205
2                25.727273              79.363636
3                88.200000              17.114286
4                26.304348              20.913043
Cluster Centroids:
Cluster 0: Annual Income = 55.30, Spending Score = 49.52
Cluster 1: Annual Income = 86.54, Spending Score = 82.13
Cluster 2: Annual Income = 25.73, Spending Score = 79.36
Cluster 3: Annual Income = 88.20, Spending Score = 17.11
Cluster 4: Annual Income = 26.30, Spending Score = 20.91
Clustered Dataset:
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   Cluster
0        4
1        2
2        4
3        2
4        4
```